TEXORA · COURSE NOTES · LECTURE SHEET

Chemical Bonding

Why atoms join and the shapes they make — ionic, covalent, coordinate and metallic bonds, hybridisation and VSEPR geometry, hydrogen bonding and the weak forces.

CHE-1105 · CHEMISTRY-I · TOPIC 2 · LEVEL 1 TERM 1

1. Why bonds form

Atoms combine because the combined state is lower in energy than the separated atoms. The classical picture of the driving force is the **octet rule**: atoms transfer or share electrons until each holds (or shares) eight electrons in its valence shell — the noble-gas arrangement. The octet is a guide, not a law (BF_3, PCl_5 and many transition-metal compounds break it), but it correctly predicts the formulas of most simple compounds and the charges of most simple ions.

2. The ionic bond

When a metal meets a non-metal of very different electronegativity, electrons are **transferred**: the metal forms a cation, the non-metal an anion, and the electrostatic attraction between the oppositely charged ions is the **ionic (electrovalent) bond**. The ions pack into a giant lattice; the energy released when the lattice forms (the **lattice energy**) is what pays for the electron transfer. Consequences follow directly: ionic solids have high melting points (the lattice must be broken), conduct electricity when molten or in solution (ions free to move) but not as solids, are often soluble in water and insoluble in organic solvents, and are brittle (a shifted layer brings like charges together and the crystal cleaves). **Fajans' rules** mark the borderline: small, highly charged cations and large, easily distorted anions polarise each other and pull the bond toward covalency — the reason $AlCl_3$ behaves far less "ionically" than NaCl.

3. The covalent bond

Atoms of similar electronegativity **share** electron pairs. A shared pair forms a single bond; two or three shared pairs form double and triple bonds, with bond length shrinking and bond strength rising along the series (C–C 154 pm, C=C 134 pm, C≡C 120 pm, approximately). Two pictures explain the sharing: **valence bond theory**, in which half-filled atomic orbitals overlap (head-on overlap gives a strong σ bond; sideways overlap of p orbitals gives the weaker π bond — a double bond is one σ + one π), and **molecular orbital theory**, in which the atoms' orbitals combine into bonding and antibonding

molecular orbitals spread over the whole molecule. MO theory earns its keep with one famous success: it predicts that O_2 has two unpaired electrons — oxygen is paramagnetic, which valence bond drawings miss entirely.

3.1 Hybridisation

To explain observed shapes, valence bond theory mixes an atom's orbitals before bonding: **sp^3** hybridisation (one s + three p) gives four tetrahedral orbitals (CH_4, 109.5°); **sp^2** gives three trigonal-planar orbitals plus one untouched p orbital for a π bond (BF_3, C_2H_4, 120°); **sp** gives two linear orbitals plus two p orbitals ($BeCl_2$, C_2H_2, 180°). Hybridisation is a model of the final geometry, not a physical event — but as a model it predicts angles and π-bond counts reliably.

3.2 The coordinate (dative) bond

A covalent bond in which **both shared electrons come from one atom** is a coordinate bond: the donor (a lone-pair holder such as NH_3 or H_2O) gives a pair to an acceptor with an empty orbital (H^+, BF_3, metal ions). NH_4^+, H_3O^+ and the $NH_3 \rightarrow BF_3$ adduct are the standard examples. Once formed, a coordinate bond is indistinguishable from any other covalent bond — NH_4^+ has four identical N–H bonds.

4. Shapes of molecules: VSEPR

The **valence shell electron pair repulsion** model makes one assumption — electron pairs around a central atom arrange themselves as far apart as possible — and from it derives the geometry of most simple molecules. Count all electron domains (bonding pairs plus lone pairs), arrange them by that count, then describe the shape by the atoms alone. Lone pairs repel more strongly than bonding pairs, so they compress bond angles (CH_4 109.5° → NH_3 107° → H_2O 104.5°).

DOMAINS	ARRANGEMENT	EXAMPLE	SHAPE OF MOLECULE / ANGLE
2	Linear	$BeCl_2$, CO_2	Linear, 180°
3	Trigonal planar	BF_3	Trigonal planar, 120°
4	Tetrahedral	CH_4	Tetrahedral, 109.5°
4 (1 lone pair)	Tetrahedral electron geometry	NH_3	Trigonal pyramidal, 107°
4 (2 lone pairs)	Tetrahedral electron geometry	H_2O	Bent, 104.5°
5	Trigonal bipyramidal	PCl_5	Trigonal bipyramidal, 120°/90°

6	Octahedral	SF_6	Octahedral, 90°

WORKED EXAMPLE — VSEPR END TO END

Predict the shape of NH_3. Nitrogen has 5 valence electrons; three form bonds to hydrogen, leaving one lone pair: 4 domains, tetrahedral electron arrangement. The atoms alone describe a trigonal pyramid. The lone pair repels more strongly than the three bonding pairs, squeezing the H–N–H angle from the ideal 109.5° down to the observed 107°. The same count for H_2O (2 bonds + 2 lone pairs) squeezes further, to 104.5°, and the shape is bent.

5. The metallic bond

In a metal, the valence electrons are not held by individual atoms but form a mobile **“sea” of electrons** around a lattice of cations. This single picture explains the metallic properties: conductivity (the sea moves under a potential), malleability (layers of cations slide while the sea holds them together — no directional bonds to snap), lustre, and high melting points for strongly bonded metals. Alloys are the same structure with different cations mixed in, usually harder because the irregular sizes obstruct sliding.

6. Between molecules: hydrogen bonds and van der Waals forces

- **Hydrogen bond** — the strong attraction between a hydrogen bonded to a small, very electronegative atom (N, O, F) and a lone pair on another such atom. It is much weaker than a covalent bond but far stronger than other intermolecular forces, and it runs everyday life: water’s high boiling point and ice’s open, floating structure, the holding of DNA’s two strands, the strength of nylon and the absorbency behaviour of cotton all trace back to it.
- **van der Waals forces** — the weak, short-range attractions between all molecules (from instantaneous dipoles), growing with molecular size and surface contact; they are why large non-polar molecules can be liquids or solids at all (iodine, waxes, polythene).
- **Dipole–dipole forces** — between permanently polar molecules, intermediate in strength.

A textile thread runs through this section: hydrogen bonding between cellulose chains is what makes cotton and paper strong when dry and weaker when wet, and what dyes and finishes must negotiate — bonding theory is never far from the mill.

7. Polarity

A bond between different electronegativities is a **polar bond** carrying partial charges (δ+ and δ−). A molecule’s overall polarity, however, depends on geometry: CO_2 has two polar bonds yet is non-polar

(the bond dipoles cancel in the linear shape), while H_2O is polar (the dipoles add in the bent shape). Molecular polarity decides solubility (“like dissolves like”), boiling point elevation among similar molecules, and how a substance behaves in an electric field and toward other chemicals.

8. Glossary — key terms of this topic

TERM	MEANING
Lattice energy	Energy released when gaseous ions combine to form one mole of an ionic solid (equivalently, needed to break the lattice).
σ and π bonds	Bonds from head-on orbital overlap (σ, strong, cylindrically symmetric) and sideways p overlap (π, weaker).
Hybridisation	The mixing of atomic orbitals into equivalent hybrid orbitals directed at the observed bond angles.
VSEPR	Valence shell electron pair repulsion theory — geometry from the mutual repulsion of electron domains.
Coordinate bond	A covalent bond in which both shared electrons are supplied by one atom.
Hydrogen bond	The attraction between a δ+ hydrogen on N, O or F and a lone pair on another N, O or F atom.
Dipole moment	The measure of a molecule’s net polarity, from partial charges and their separation.

9. Key points and summary

- Bonds form to lower energy; transfer gives ionic lattices, sharing gives covalent molecules, delocalisation gives metals — and Fajans’ rules mark the ionic–covalent borderland.
- Valence bond theory (overlap, σ/π, hybridisation) and MO theory both describe sharing; MO theory alone predicts O_2’s paramagnetism.
- VSEPR turns an electron-domain count into geometry; lone pairs compress angles (109.5° → 107° → 104.5°).
- Between molecules, hydrogen bonds and van der Waals forces set boiling points, solubility and the wet behaviour of fibres; molecular polarity is bond polarity filtered through geometry.

10. Practice

Answer in your own words; brief model answers follow.

1. State the bond type in each and one property that follows: NaCl, CH_4, Cu, NH_4^+ (for the N–H bond formed last).
2. Using VSEPR, predict the shape and approximate bond angle of (a) CO_2, (b) BF_3, (c) H_2S (treat S like O).
3. Why is the H–O–H angle (104.5°) smaller than the H–N–H angle (107°), which is smaller than the H–C–H angle (109.5°)?
4. Carbon dioxide has polar bonds but is a non-polar molecule; water is polar. Explain both facts with geometry.
5. State the hybridisation of carbon in (a) CH_4, (b) C_2H_4, (c) C_2H_2, and the number of π bonds in each molecule.
6. Ice floats on water, and water boils far higher than H_2S. Name the single interaction responsible for both and describe it.
7. By Fajans' rules, which is more covalent in character, NaCl or $AlCl_3$? Give the two reasons.
8. MO theory's famous success: state the observation about oxygen it predicts that simple valence-bond drawings do not.

Answer key

1. NaCl — ionic (high melting point, conducts when molten). CH_4 — covalent (low melting/boiling, non-conductor). Cu — metallic (conducts as a solid, malleable). NH_4^+'s fourth N–H bond — coordinate (dative), formed from NH_3's lone pair and H^+.
2. (a) Linear, 180° (2 domains). (b) Trigonal planar, 120° (3 domains). (c) Bent, about 104.5° (4 domains, 2 lone pairs).
3. Each step adds a lone pair, and lone pairs repel more strongly than bonding pairs, compressing the remaining bond angle further.
4. In linear CO_2 the two bond dipoles are equal and opposite, so they cancel. In bent H_2O the two dipoles point partly in the same direction and add, giving a net dipole.
5. (a) sp^3, 0 π bonds. (b) sp^2, 1 π bond (the double bond = σ + π). (c) sp, 2 π bonds (the triple bond = σ + 2π).
6. Hydrogen bonding — the attraction between δ+ H on oxygen and lone pairs of neighbouring oxygens; it holds ice in an open lattice (less dense) and must be broken to boil water.
7. $AlCl_3$ — Al^{3+} is smaller and more highly charged than Na^+, so it polarises the large Cl^- electron cloud far more strongly.
8. Oxygen is paramagnetic — O_2 has two unpaired electrons (in antibonding π orbitals), which the MO diagram predicts and attracts a magnet, as observed.